Web Technology Programs

UNIVERSITY OF JAMMU

HOME LISTS TABLES IMAGES

DEPARTMENT OF COMPUTER SCIENCE &IT

Submitted by

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MCA 3RD Sem Subjects

- **Web Technologies**
- HTML/CSS/XML
- JavaScript
- Servlets and JSP
- ASP Controls
- State Management in ASP
- **Machine Learning Using Python**
- Introduction to Machine Learning
- Types of Machine Learning
- Genetic Algorithm and Deep Learning
- Introduction to Python
- Introduction to Machine Learning using Python and Deep Learning
- **Theory of Computation and Compiler Design**
- Regular Languages and Expressions
- Context Free Grammar
- Turing Machines
- Compiler Structure & Frontend
- Errors Detection and Recovery, Code Generation & Optimization

MCA 3rd Semester Time Table

Day	10:00 AM - 11:00 AM	11:00 AM - 12:00 PM	12:00 PM - 1:00 PM	1:00 PM - 1:30 PM	1:30 PM - 2:30 PM	2:30 PM - 4:30 PM
Monday	Web Technologies	Software Engineering	Theory of Computation and compiler design	Break	Machine learning using Python	Practical
Tuesday	Web Technologies	Software Engineering	Theory of Computation and compiler design	Break	Big data analytics using R	Practical
Wednesday	Big data analytics using R	Theory of Computation and compiler design	Software Engineering	Break	Machine learning using Python	Practical
Thursday	Web Technologies	Big data analytics using R	Theory of Computation and compiler design	Break	Machine learning using Python	Practical
Friday	Web Technologies	Software Engineering	Machine learning using Python	Break	Big data analytics using R	Practical

Teaching/Non-Teaching Record Table

Faculty Name	Designation	Area of Interest
Dr. Shubhnandan Singh Jamwal	Professor	Java & Web Technologies
Sh. Jasbir Singh	Assistant Professor	Theory of Computation & ADA
Sh. Anil Tickoo	Programmer	C++ & Python
Sh. Sanjay Manhas	Technical Staff	Workshop
Mr. Ajay Malgotra	Non-Teaching Staff	Department

Images

1. Design front static page for department:

```
<!DOCTYPE html>
<html lang="en">

<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Department of Computer Science & IT - University of Jammu</title>
<style>
@import url('https://fonts.googleapis.com/css2?family=Jaro:opsz@6..72&family=Poppins:ital,wght@0,100;0,200;0,300;0,400;0,500;0,600;0,700;0,800;0,900;1,100;1,200;1,300;1,400;1,500;1,600;1,700;1,800;1,900&display=swap');
*{
font-family: 'Poppins',Arial, sans-serif;
margin: 0;
padding: 0 1px;
}

header {
background-color: #f4f4f4;
padding: 7px 0;
text-align: center;
border-radius: 7px;
margin-bottom: 2px;
}
header img{
width: 300px;
border-radius: 12px;
}

nav {
background-color: #f4f4f4;
padding: 16px;
text-align: center;
margin-bottom: 7px;
}

nav a {
margin: 0 20px;
text-decoration: none;
color: #6d6dd2;
font-weight: bold;
}

nav a:hover {
color: #ff6600;
border-bottom: 3px solid #ff6600;
padding-bottom: 15px;
}
nav a:active{
color: #ff6600;
border-bottom: 3px solid #ff6600;
padding-bottom: 15px;
}

h1{
text-align: center;
color: #060f12;
font-family: "Jaro",Arial, sans-serif;
}
h3{
color: #6885a7;
}
li{
margin: 2px 17px;
}
.student-details{
background-color: #f4f4f4;
text-align: center;
line-height: 1.5;
}
.syllabus-part, .semester-div {
background-color: #f4f4f4;
margin: 20px 0;
padding: 7px 30px;
max-width: auto;
}
.syllabus-part h3{
text-align: center;
}

table {
width: 100%;
border-collapse: collapse;
margin: 20px 0;
}

table,
```

```
th,
td {
border: 1px solid #ddd;
}

th,
td {
padding: 7px;
text-align: left;
}

th {
background-color: #6d6dd2;
color: white;
}
.dept-imgs img{
width: 240px;
display: block;
margin: 10px auto;
}
</style>
</head>

<body>
<header>
<img
src="https://www.jammuuniversity.ac.in/sites/d
efault/files/logo.png" alt="University Logo">
</header>
<nav>
<a href="#">HOME</a>
<a href="#">LISTS</a>
<a href="#">TABLES</a>
<a href="#">IMAGES</a>
</nav>
<main>
<h1><span>DEPARTMENT OF COMPUTER
SCIENCE &IT </span></h1>
<div class="student-details">
<h3>Submitted by ⤵️ </h3>
<p>Himanshu Parihar | 30-MCA-23</p>
<p>Agrim Sharma | 16-MCA-23</p>
<p>Kapil Sadotra | 13-MCA-23</p>
<p>Atul Kohli | 11-MCA-23</p>
<p>Prabhjeet Choudhary | 20-MCA-23</p>
</div>
<div class="syllabus-part">
<h3>MCA 3<sup>RD</sup> Sem Subjects</h3>
```

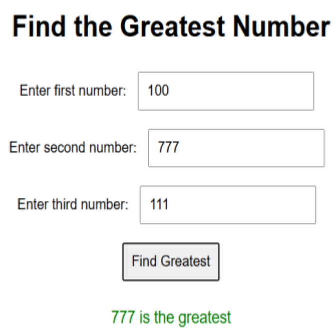
```
<ul>
<li><strong>Web Technologies</strong>
<ul>
<li>HTML/CSS/XML</li>
<li>JavaScript</li>
<li>Servlets and JSP</li>
<li>ASP Controls</li>
<li>State Management in ASP</li>
</ul>
</li>
<li><strong>Machine Learning Using
Python</strong>
<ul>
<li>Introduction to Machine
Learning</li>
<li>Types of Machine Learning</li>
<li>Genetic Algorithm and Deep
Learning</li>
<li>Introduction to Python</li>
<li>Introduction to Machine Learning
using Python and Deep Learning</li>
</ul>
</li>
<li><strong>Theory of Computation and
Compiler Design</strong>
<ul>
<li>Regular Languages and
Expressions</li>
<li>Context Free Grammar</li>
<li>Turing Machines</li>
<li>Compiler Structure & Frontend</li>
<li>Errors Detection and Recovery,
Code Generation & Optimization</li>
</ul>
</li>
</ul>
</li>
</ul>
</div>
<div class="semester-div">
<h3>MCA 3rd Semester Time Table</h3>
<table>
<thead>
<tr>
<th>Day</th>
<th>10:00 AM - 11:00 AM</th>
<th>11:00 AM - 12:00 PM</th>
<th>12:00 PM - 1:00 PM</th>
```

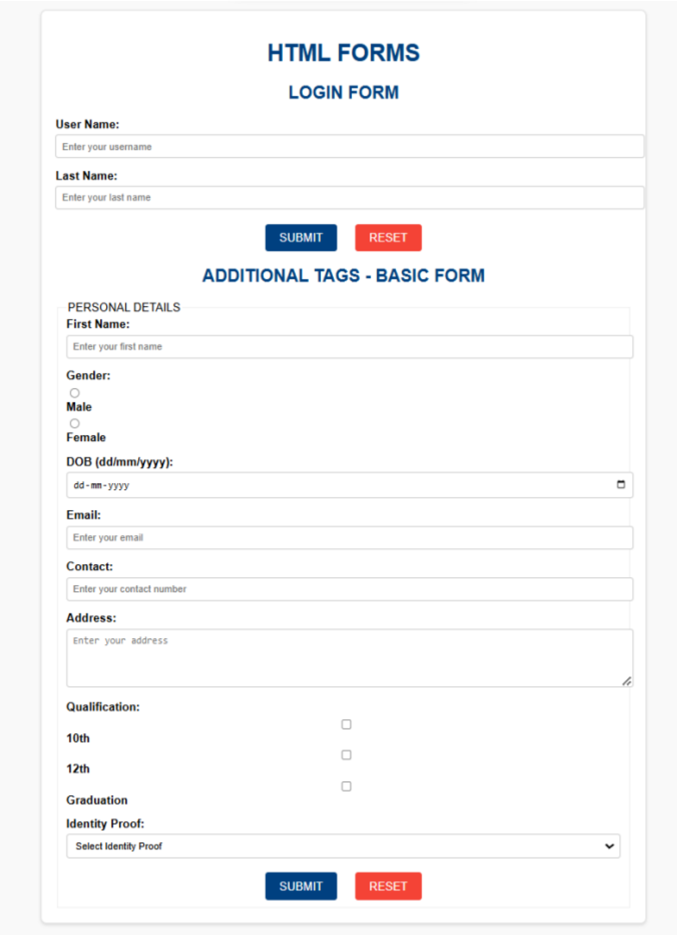
```
<th>1:00 PM - 1:30 PM</th>
<th>1:30 PM - 2:30 PM</th>
<th>2:30 PM - 4:30 PM</th>
</tr>
</thead>
<tbody>
<tr>
<td>Monday</td>
<td>Web Technologies</td>
<td>Software Engineering</td>
<td>Theory of Computation and compiler
design</td>
<td>Break</td>
<td>Machine learning using Python</td>
<td>Practical</td>
</tr>
<tr>
<td>Tuesday</td>
<td>Web Technologies</td>
<td>Software Engineering</td>
<td>Theory of Computation and
compiler design</td>
<td>Break</td>
<td>Big data analytics using R</td>
<td>Practical</td>
</tr>
<tr>
<td>Wednesday</td>
<td>Big data analytics using R</td>
<td>Theory of Computation and
compiler design</td>
<td>Software Engineering</td>
<td>Break</td>
<td>Machine learning using Python</td>
<td>Practical</td>
</tr>
<tr>
<td>Thursday</td>
<td>Web Technologies</td>
<td>Big data analytics using R</td>
<td>Theory of Computation and
compiler design</td>
<td>Break</td>
<td>Machine learning using Python</td>
<td>Practical</td>
</tr>
<tr>
<td>Friday</td>
<td>Web Technologies</td>
<td>Software Engineering</td>
<td>Machine learning using Python</td>
<td>Break</td>
<td>Big data analytics using R</td>
<td>Practical</td>
</tr>
</tbody>
</table>
<h3>Teaching/Non-Teaching Record
Table</h3>
<table>
<thead>
<tr>
<th>Faculty Name</th>
<th>Designation</th>
<th>Area of Interest</th>
</tr>
</thead>
<tbody>
<tr>
<td>Dr. Shubhnandan Singh Jamwal</td>
<td>Professor</td>
<td>Java & Web Technologies</td>
</tr>
<tr>
<td>Sh. Jasbir Singh</td>
<td>Assistant Professor</td>
<td>Theory of Computation & ADA</td>
</tr>
<tr>
<td>Sh. Anil Tickoo</td>
<td>Programmer</td>
<td>C++ & Python</td>
</tr>
<tr>
<td>Sh. Sanjay Manhas</td>
<td>Technical Staff</td>
<td>Workshop</td>
</tr>
<tr>
<td>Mr. Ajay Malgotra</td>
<td>Non-Teaching Staff</td>
<td>Department</td>
</tr>
</tbody>
</table>
</div>
```

```html
<div class="dept-imgs">
<h3>Images</h3>
<img src="./studentsx1.jpg" alt="Department Picture">
</div>
</main>
</body>
</html>
```

2. WAP to find the greatest of three numbers (user defined input) using ternary operators:

```html
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Find Greatest Number</title>
<style>
body {
font-family: Arial, sans-serif;
text-align: center;
margin-top: 50px;
}
input, button {
margin: 10px;
padding: 10px;
font-size: 16px;
}
#output {
margin-top: 20px;
font-size: 18px;
color: green;
}
</style>
</head>
<body>
<h1>Find the Greatest Number</h1>
<label for="num1">Enter first number:</label>
<input type="number" id="num1" placeholder="First Number"><br>
<label for="num2">Enter second number:</label>
<input type="number" id="num2" placeholder="Second Number"><br>
<label for="num3">Enter third number:</label>
<input type="number" id="num3" placeholder="Third Number"><br>
<button onclick="findGreatest()">Find Greatest</button>
<div id="output"></div>
<script>
function greatest(a, b, c) {
let value = a > b && a > c ? a
: b > a && b > c ? b
: c;
return value;
}
function findGreatest() {
let num1 = document.getElementById('num1').value;
let num2 = document.getElementById('num2').value;
let num3 = document.getElementById('num3').value;
let result = greatest(num1, num2, num3);
document.getElementById('output').innerText = `${result} is the greatest`;
}
</script>
</body>
</html>
```





3. Write HTML code to create login form and additional tags form as shown in the figure using the div and required tags:

```
<!DOCTYPE html>
<html lang="en">

<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Practical 2</title>
<style>
body {
font-family: Arial, sans-serif;
margin: 0;
padding: 0;
background-color: #f9f9f9;
}

.container {
width: 90%;
max-width: 800px;
margin: 20px auto;
padding: 20px;
background-color: #ffffff;
border: 1px solid #ddd;
border-radius: 8px;
box-shadow: 0 2px 4px rgba(0, 0, 0, 0.1);
}

h1,
h2 {
text-align: center;
color: #004080;
}

form {
margin-top: 20px;
}

.form-group {
margin-bottom: 15px;
}

.form-group label {
display: block;
margin-bottom: 5px;
font-weight: bold;
}

.form-group input,
.form-group select,
.form-group textarea {
width: 100%;
padding: 8px;
border: 1px solid #ccc;
border-radius: 4px;
}

.form-group input[type="radio"] {
width: auto;
}

.form-actions {
```

```
text-align: center;
margin-top: 20px;
}

.form-actions button {
padding: 10px 20px;
margin: 0 10px;
border: none;
border-radius: 4px;
cursor: pointer;
font-size: 16px;
}

.form-actions button.submit {
background-color: #004080;
color: white;
}

.form-actions button.reset {
background-color: #f44336;
color: white;
}
</style>
</head>

<body>
<div class="container">
<h1>HTML FORMS</h1>
<h2>LOGIN FORM</h2>
<form>
<div class="form-group">
<label for="username">User Name:</label>
<input type="text" id="username" name="username" placeholder="Enter your username" required>
</div>
<div class="form-group">
<label for="lastname">Last Name:</label>
<input type="text" id="lastname" name="lastname" placeholder="Enter your last name" required>
</div>
<div class="form-actions">
<button type="submit" class="submit">SUBMIT</button>
<button type="reset" class="reset">RESET</button>
</div>
</form>
<h2>ADDITIONAL TAGS - BASIC FORM</h2>
<form>
<fieldset>
<legend>PERSONAL DETAILS</legend>
<div class="form-group">
<label for="firstname">First Name:</label>
<input type="text" id="firstname" name="firstname" placeholder="Enter your first name" required>
</div>
<div class="form-group">
<label>Gender:</label>
<input type="radio" id="male" name="gender" value="Male" required>
<label for="male">Male</label>
<input type="radio" id="female" name="gender" value="Female">
<label for="female">Female</label>
</div>
<div class="form-group">
<label for="dob">DOB (dd/mm/yyyy):</label>
<input type="date" id="dob" name="dob" required>
</div>
<div class="form-group">
<label for="email">Email:</label>
<input type="email" id="email" name="email" placeholder="Enter your email" required>
</div>
<div class="form-group">
<label for="contact">Contact:</label>
<input type="tel" id="contact" name="contact" placeholder="Enter your contact number" required>
</div>
<div class="form-group">
<label for="address">Address:</label>
<textarea id="address" name="address" rows="4" placeholder="Enter your address"></textarea>
</div>
<div class="form-group">
```

```
<label>Qualification:</label>
<input type="checkbox" id="qualification10"
name="qualification" value="10th">
<label for="qualification10">10th</label>
<input type="checkbox" id="qualification12"
name="qualification" value="12th">
<label for="qualification12">12th</label>
<input type="checkbox" id="graduation"
name="qualification" value="Graduation">
<label for="graduation">Graduation</label>
</div>
<div class="form-group">
<label for="identity">Identity
Proof:</label>
<select id="identity" name="identity" required>
<option value="">Select Identity
Proof</option>
<option value="aadhar">Aadhar
Card</option>
<option value="pan">PAN Card</option>
<option value="passport">Passport</option>
</select>
</div>
<div class="form-actions">
<button type="submit"
class="submit">SUBMIT</button>
<button type="reset"
class="reset">RESET</button>
</div>
</fieldset>
</form>
</div>
</body>
</html>
```

4. Define external styling sheet:

   a. *To style paragraph using id names "para1".*

Styles.css file

```
#para1{
color: green;
}
```

b. Add stylesheet (internal/ external) in Question 1.

Internal Stylesheet –

Already used with <style> tag

External Stylesheet –

Style.css

```
*{
font-family: 'Poppins',Arial, sans-serif;
margin: 0;
padding: 0 1px;
}

header {
background-color: #f4f4f4;
padding: 7px 0;
text-align: center;
border-radius: 7px;
margin-bottom: 2px;
}
header img{
width: 300px;
border-radius: 12px;
}

nav {
background-color: #f4f4f4;
padding: 16px;
text-align: center;
margin-bottom: 7px;
}

nav a {
margin: 0 20px;
text-decoration: none;
color: #6d6dd2;
font-weight: bold;
}

nav a:hover {
color: #ff6600;
border-bottom: 3px solid #ff6600;
padding-bottom: 15px;
}
nav a:active{
color: #ff6600;
border-bottom: 3px solid #ff6600;
padding-bottom: 15px;
}

h1{
```

```
text-align: center;
color: #060f12;
font-family: "Jaro",Arial, sans-serif;
}
h3{
color: #6885a7;
}
li{
margin: 2px 17px;
}
.student-details{
background-color: #f4f4f4;
text-align: center;
line-height: 1.5;
}
.syllabus-part, .semester-div {
background-color: #f4f4f4;
margin: 20px 0;
padding: 7px 30px;
max-width: auto;
}
.syllabus-part h3{
text-align: center;
}

table {
width: 100%;
border-collapse: collapse;
margin: 20px 0;
}

table,
th,
td {
border: 1px solid #ddd;
}

th,
td {
padding: 7px;
text-align: left;
}
th {
background-color: #6d6dd2;
color: white;
}
.dept-imgs img{
width: 240px;
display: block;
margin: 10px auto;
}
```

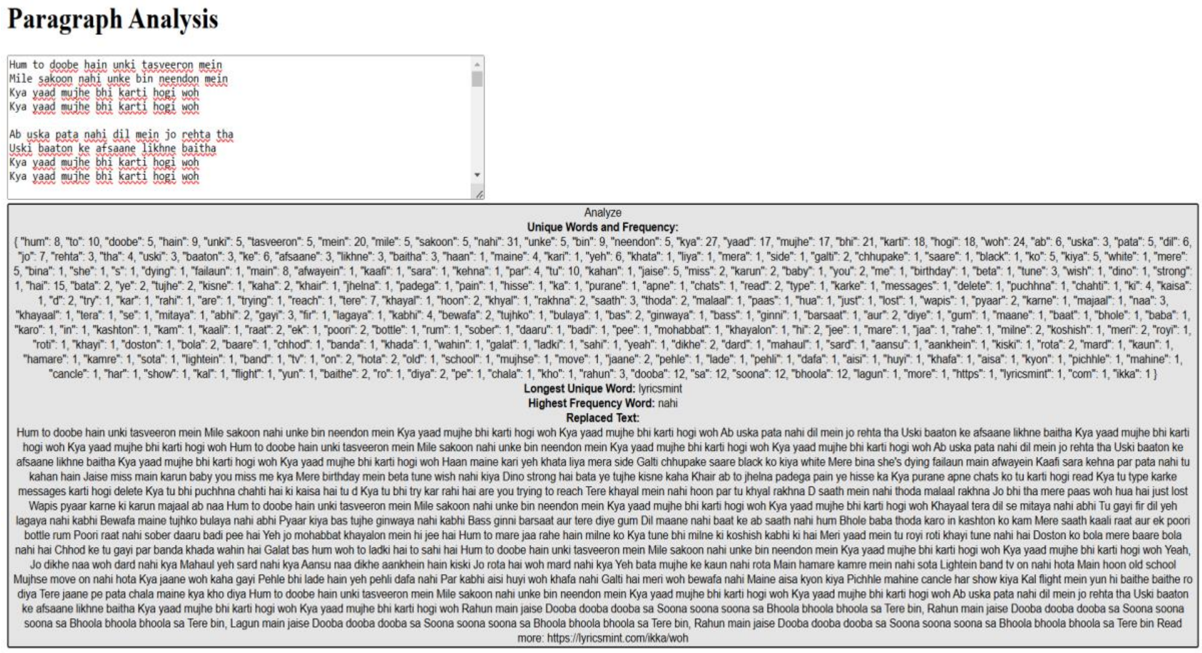
5. WAP to Implement the Methods of Math Class:

```
<!DOCTYPE html>
<html lang="en">

<head>
<meta charset="UTF-8">
<meta name="viewport"
content="width=device-width, initial-
scale=1.0">
<title>Math Class Methods</title>
<script>
function showMathMethods() {
let num = 25.6;
let result = `
Math.ceil(${num}) =
${Math.ceil(num)}<br>
Math.floor(${num}) =
${Math.floor(num)}<br>
Math.round(${num}) =
${Math.round(num)}<br>
Math.sqrt(16) = ${Math.sqrt(16)}<br>
Math.pow(2, 3) = ${Math.pow(2, 3)}<br>
Math.abs(-5) = ${Math.abs(-5)}<br>
Math.min(3, 7, 1) = ${Math.min(3, 7,
1)}<br>
Math.max(3, 7, 1) = ${Math.max(3, 7,
1)}<br>
Math.random() =
${Math.random().toFixed(2)}<br>`;
document.write(result)
}
</script>
</head>

<body>
<h1>Math Class Methods</h1>
<button onclick="showMathMethods()">Show
Results</button>
<div id="mathResults"></div>
</body>

</html>
```

```
Math.ceil(25.6) = 26
Math.floor(25.6) = 25
Math.round(25.6) = 26
Math.sqrt(16) = 4
Math.pow(2, 3) = 8
Math.abs(-5) = 5
Math.min(3, 7, 1) = 1
Math.max(3, 7, 1) = 7
Math.random() = 0.14
```

6. Paragraph Analysis with Frequency, Unique Words, and Replacement:

```
<!DOCTYPE html>
<html lang="en">

<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Paragraph Analysis</title>
<script>
function analyzeText() {
let text = document.getElementById("inputText").value;
let words = text.toLowerCase().match(/\b\w+\b/g);
let frequency = {};
let longestUniqueWord = "";
let highestFrequencyWord = "";
let maxFrequency = 0;
words.forEach(word => {
frequency[word] = (frequency[word] || 0) + 1;
if (word.length > longestUniqueWord.length && frequency[word] === 1) {
longestUniqueWord = word;
}
if (frequency[word] > maxFrequency) {
maxFrequency = frequency[word];
highestFrequencyWord = word;
}
});
let replacedText = text.replace(/\ba\b/gi, "ComputerScienceDepartmentUniversityOf Jammu");
document.getElementById("output").innerHTML = `<strong>Unique Words and Frequency:</strong><br>${JSON.stringify(frequency, null, 2)} <br>
<strong>Longest Unique Word:</strong> ${longestUniqueWord}<br>
<strong>Highest Frequency Word:</strong> ${highestFrequencyWord}<br>
<strong>Replaced Text:</strong><br>${replacedText}<br>`;
}
</script>
</head>

<body>
<h1>Paragraph Analysis</h1>
<textarea id="inputText" rows="10" cols="80" placeholder="Enter your paragraph here..."></textarea>
<br>
<button onclick="analyzeText()">Analyze</button>
<div id="output"></div>
</body>
</html>
```



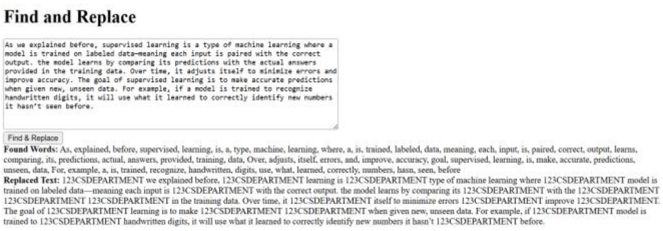
7. Gmail Registration form with validation:

```
<!DOCTYPE html>
<html lang="en">

<head>
<meta charset="UTF-8">
<meta name="viewport"
content="width=device-width, initial-
scale=1.0">
<title>Gmail Registration Form</title>
<style>
body{
padding: 13px 13px;
}
input{
margin: 7px 0;
}
</style>
<script>
function validateForm() {
const fname =
document.getElementById("fname").value.trim(
);
const lname =
document.getElementById("lname").value.trim(
);
const email =
document.getElementById("email").value.trim()
;
const password =
document.getElementById("password").value.tr
im();
const confirmPassword =
document.getElementById("confirmPassword")
.value.trim();
if (!fname || !lname || !email ||!password ||
!confirmPassword) {
alert("All fields are required.");
return false;
}
if (password !== confirmPassword) {
alert("Passwords do not match.");
return false;
}
const emailPattern = /^[a-zA-Z0-9._-]+@[a - zA -
Z0 - 9. -] +\.[a - zA - Z]{ 2, 6 } $ /;
if (!emailPattern.test(email)) {
alert("Invalid email format.");
return false;
}
alert("Registration Successful!");
return true;
}
</script>
</head>
```

```
<body>
<h1>Gmail Registration Form</h1>
<form onsubmit="return validateForm()">
First Name: <input type="text" id="fname"><br>
Last Name: <input type="text" id="lname"><br>

Email: <input type="email" id="email"><br>

Password: <input type="password"
id="password"><br>


Confirm Password: <input type="password"
id="confirmPassword"><br>
<button type="submit">Submit</button>
</form>
</body>

</html>
```

8. Find and Replace Operation:

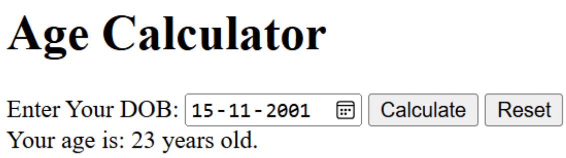
```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8" />
<meta name="viewport"
content="width=device-width, initial scale=1.0"
/>
<title>Find and Replace</title>
<script>
function findReplace() {
let text =
document.getElementById("inputText").value;
let words = text.match(/\b\w+\b/g) || [];
let foundWords = words.filter(word =>
/[aprs]/i.test(word));
let replacedText = text.replace(/\b[aprs]\w*/gi,
"123CSDEPARTMENT");
document.getElementById("output").innerHTM
L = `
<strong>Found Words:</strong>
${foundWords.join(", ")}<br>
<strong>Replaced Text:</strong>
${replacedText}`;
}
</script>

</head>
<body>
<h1>Find and Replace</h1>
<textarea id="inputText" rows="10" cols="80"
placeholder="Enter your text
here..."></textarea>
<br/>
<button onclick="findReplace()">Find &
Replace</button>
<div id="output"></div>
</body>
</html>
```



9. Age Calculations with DOB:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport"
content="width=device-width, initial-
scale=1.0">
<title>Age Calculator</title>
</head>
<body>
<h1>Age Calculator</h1>
<form>
<label for="dob">Enter Your DOB:</label>
<input id="dob" type="date">

<button type="button" id="cal-
btn">Calculate</button>
<button type="reset" id="rst-
btn">Reset</button>

<div id="age-result"></div>
</form>

<script>
let dob = document.getElementById('dob');
let calculate = document.getElementById('cal-
btn');
let ageResult = document.getElementById('age-
result');
let rstbtn = document.getElementById('rst-btn');

calculate.addEventListener('click', function() {
if (dob.value) {
let birthDate = new Date(dob.value);
let currentDate = new Date();
let age = currentDate.getFullYear() -
birthDate.getFullYear();
let monthDiff = currentDate.getMonth() -
birthDate.getMonth();

if (monthDiff < 0 || (monthDiff === 0 &&
currentDate.getDate() < birthDate.getDate())) {
age--;
}
```

```
ageResult.textContent = `Your age is: ${age}
years old.`;
} else {
ageResult.textContent = 'Please select a valid
date of birth.';
}
});
</script>

</body>
</html>
```



R Programs

1. WAP to display any value in vector from location:

```
my_vector <- c(10, 20, 30, 40, 50)
 index <- 3
 value_at_index <- my_vector[index]
 print(value_at_index)
```

```
> source("c:\\Users\\hp\\.anaconda\\R PRograms\\R1.R", encoding = "UTF-8")
[1] 30
>
```

2. WAP to add two numbers using functions.

```
twoNumAdder <- function(num1, num2){
 return(num1 + num2)
}
number1 <- as.integer(readline(prompt = "Enter first number: "))
number2 <- as.integer(readline(prompt = "Enter second number: "))
paste("The sum is ",twoNumAdder(number1, number2))
```

```
Enter first number: 5
Enter second number: 10
[1] "The sum is  15"
```

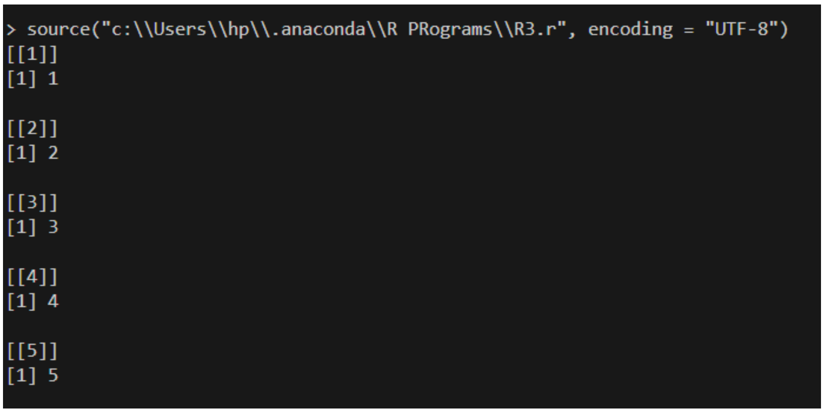
3. WAP to check if a number is even or odd by using function.

```
is_even <- function(num) {
  if (num %% 2 == 0) {
    return("Even")
  } else {
    return("Odd")
  }
}
number <- 25
result <- is_even(number)
print(result)
```

```
> source("c:\\Users\\hp\\.anaconda\\R PRograms\\r2.r", encoding = "UTF-8")
[1] "Odd"
>
```

4. Create a list, add an element, and remove the last element.

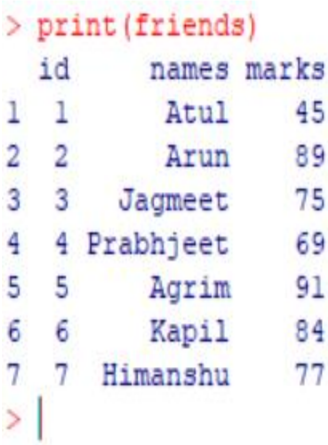
```
my_list <- list(1, 2, 3, 4, 5)
my_list <- append(my_list, 6)
my_list <- my_list[-length(my_list)]
print(my_list)
```



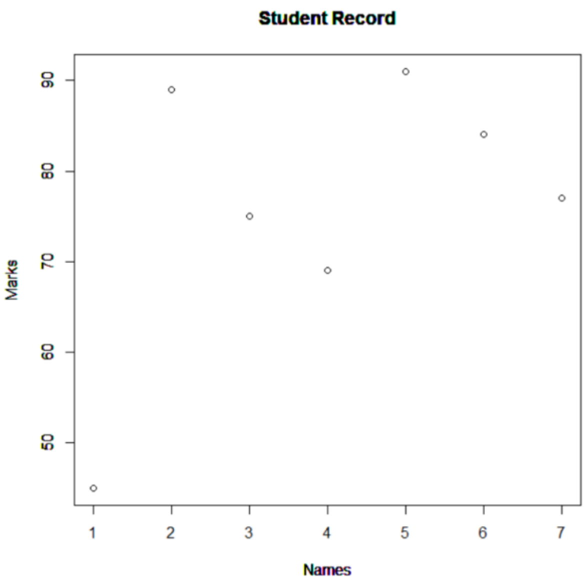
```
> source("c:\\Users\\hp\\.anaconda\\R PRograms\\R3.r", encoding = "UTF-8")
[[1]]
[1] 1

[[2]]
[1] 2

[[3]]
[1] 3

[[4]]
[1] 4

[[5]]
[1] 5
```

5. To create a data frame. Display plot, bar-plot and pie chart.

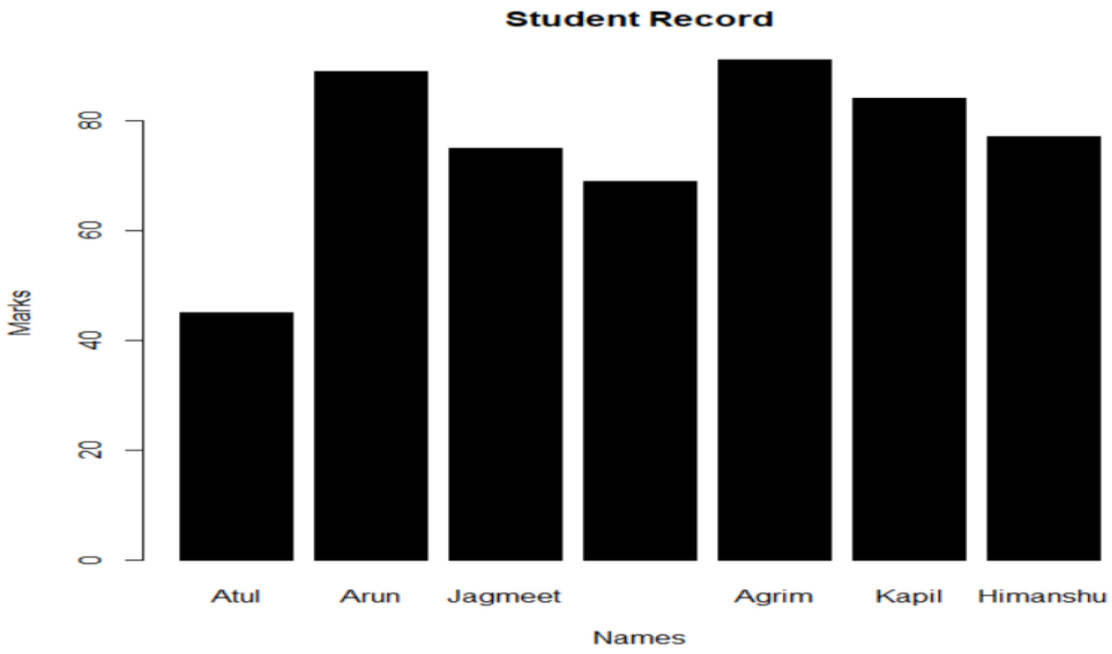
```
friends <- data.frame(
 id = c(1:7),
 names = c("Atul","Arun","Jagmeet","Prabhjeet","Agrim","Kapil","Himanshu"),
 marks = c(45,89,75,69,91,84,77)
 )
 print(friends)
 plot(friends$marks, xlab="Names",ylab="Marks", main="Student Record")
 barplot(friends$marks, xlab="Names",ylab="Marks", main="Student Rec
ord", names.arg=friends$names,col="black")
 pie(friends$marks, labels=friends$names,main="Student Record")
```

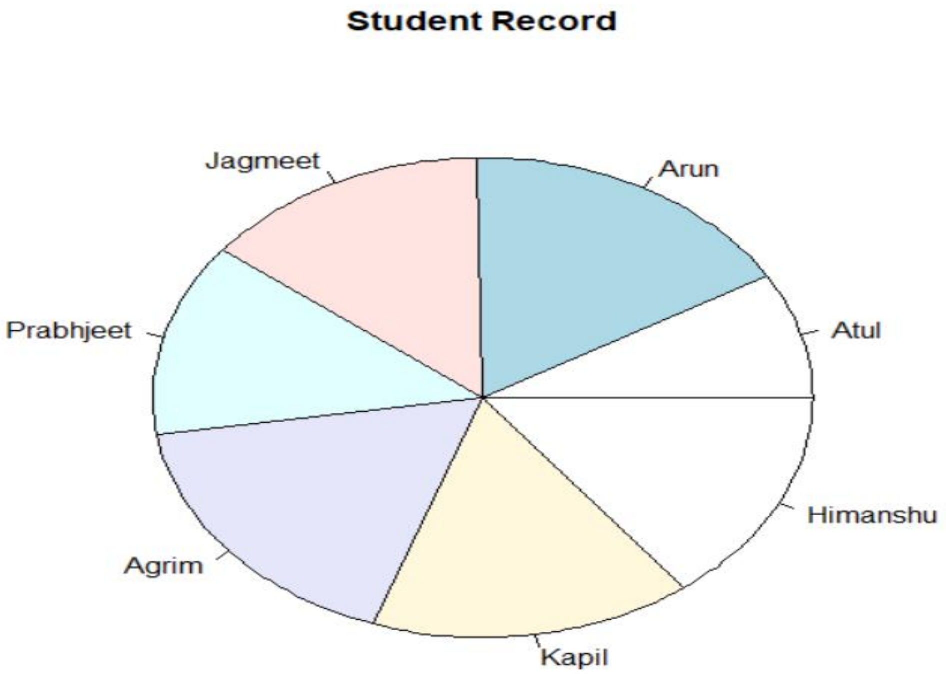


```
> print(friends)
  id     names marks
1  1      Atul    45
2  2      Arun    89
3  3   Jagmeet    75
4  4 Prabhjeet    69
5  5     Agrim    91
6  6     Kapil    84
7  7  Himanshu    77
> 
```



'





6. WAP to create data frame for 10 students containing rollno, name & marks of 3 subjects - sub1, sub2 & sub3.

```
df <- data.frame(
Roll_No = 1:10,
```

```
   Name = c("Agrim", "Bobby", "Prabjeet", "Mansi", "Himanshu", "Atul", "Etika",
"Mona", "Raj", "Kapil"),

   Sub1 = c(85, 92, 78, 95, 88, 79, 90, 82, 93, 87),

   Sub2 = c(90, 88, 92, 85, 91, 86, 93, 89, 87, 94),

   Sub3 = c(88, 91, 87, 90, 89, 85, 92, 88, 90, 86)

)

print(df)
```

```
> source("c:\\Users\\hp\\.anaconda\\R PRograms\\r4.r", encoding = "UTF-8")
   Roll_No     Name Sub1 Sub2 Sub3
1        1    Agrim   85   90   88
2        2    Bobby   92   88   91
3        3 Prabjeet   78   92   87
4        4    Mansi   95   85   90
5        5 Himanshu   88   91   89
6        6     Atul   79   86   85
7        7    Etika   90   93   92
8        8     Mona   82   89   88
9        9      Raj   93   87   90
10      10    Kapil   87   94   86
>
```

7. Generate dataset with 10 rows and 4 columns(ID, Name, Age, Score). Include atleast 5 NA values distributed across different columns. Print the original dataset.

```
df <- data.frame(

 ID = 1:10,

Name = c("Agrim", "Himanshu", "Prabjeet", "Mansi", "Manisha", "Atul", "Etika",
"Mona", "Raj", "Kapil"),

Age = c(20, 22, 21, NA, NA, 24, 25, 26, NA, 23),

 Score = c(85, 92, 78, 78, NA, 99, NA, NA, 82, 86)

)

print(df)
```

	ID	Name	Age	Score
1	1	Agrim	20	85
2	2	Himanshu	22	92
3	3	Prabjeet	21	78
4	4	Mansi	NA	78
5	5	Manisha	NA	NA
6	6	Atul	24	99
7	7	Etika	25	NA
8	8	Mona	26	NA
9	9	Raj	NA	82
10	10	Kapil	23	86

Write Code to:

a. Remove all rows that contain NA values from the dataset and print the cleaned dataset.

```
df_cleaned <- na.omit(df)

print(df_cleaned)
```

	ID	Name	Age	Score
1	1	Agrim	20	85
2	2	Himanshu	22	92
3	3	Prabjeet	21	78
6	6	Atul	24	99
10	10	Kapil	23	86

b. Replace NA values in numeric columns (Age & Score) with median of the respective column and print the dataset

```
median_age <- median(df$Age, na.rm = TRUE)
```

median_score <- median(df$Score, na.rm = TRUE)

df$Age[is.na(df$Age)] <- median_age

df$Score[is.na(df$Score)] <- median_score

print(df)

```
   ID     Name Age Score
1   1    Agrim  20    85
2   2 Himanshu  22    92
3   3 Prabjeet  21    78
4   4    Mansi  23    78
5   5  Manisha  23    85
6   6    Atul   24    99
7   7   Etika   25    85
8   8    Mona   26    85
9   9     Raj   23    82
10 10   Kapil   23    86
```

8. Perform slice and filter operation on dataset of program

Slice Operation

sliced_data <- df[3:6, c("Name", "Score")]

print(sliced_data)

```
      Name Score
3 Prabjeet    78
4    Mansi    78
5  Manisha    NA
6     Atul    99
```

Filter Operation

filtered_data <- df[df$Score >= 85,]

print(filtered_data)

	ID	Name	Age	Score
1	1	Agrim	20	85
2	2	Himanshu	22	92
3	3	Prabjeet	21	88
6	6	Atul	24	99
10	10	Kapil	23	86

9. From the given data set:
    a. Calculate the covariance
    b. Calculate the correlation coefficient
    c. Interpret the results

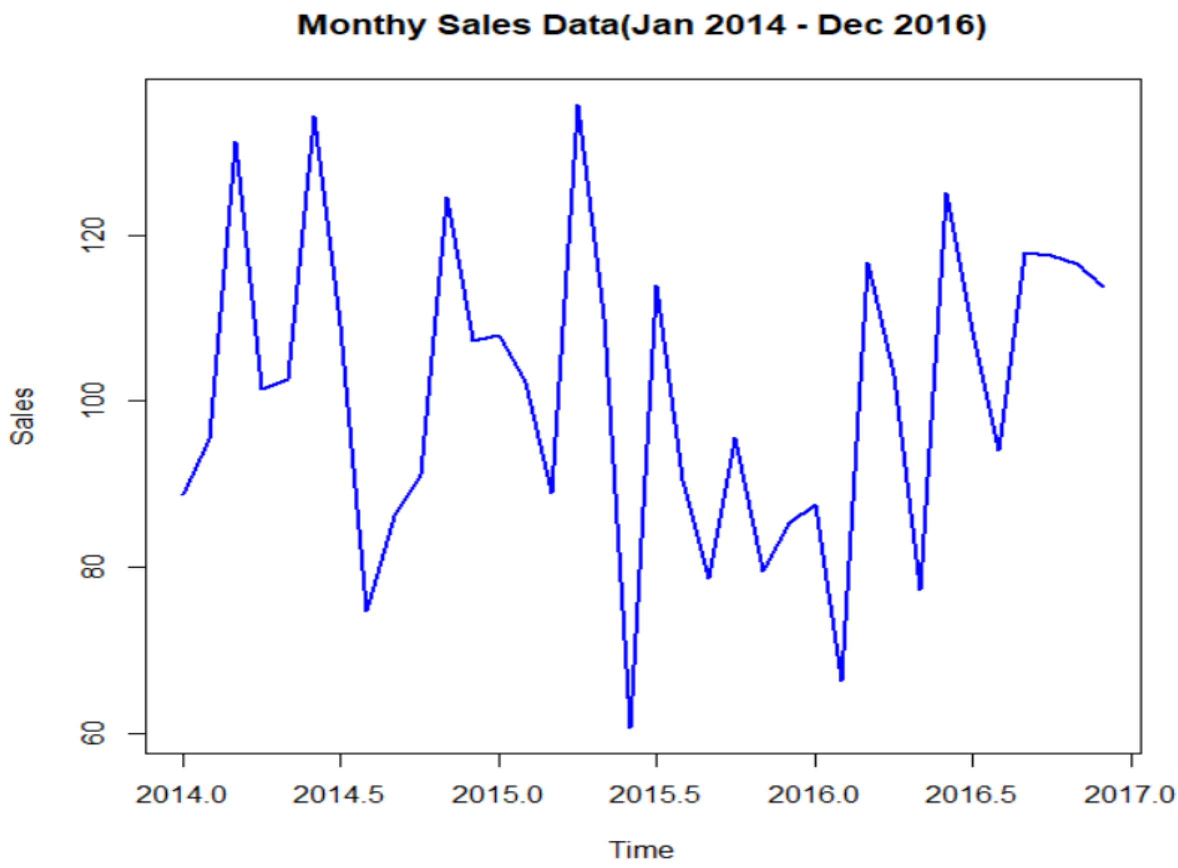
```
study_hours <- c(2,4,6,8,10)
exam_scores <- c(50,55,60,70,85)
covariance <- cov(study_hours, exam_scores)


correlation_coefficient <- cor(study_hours, exam_scores)
covariance
correlation_coefficient
```

```
>
> covariance
[1] 42.5
> correlation_coefficient
[1] 0.9686649
>
```

10. Create a time series object.

```
set.seed(123)
sales_data <- rnorm(36, mean=100, sd=20)
```

```
sales_ts <- ts(sales_data, start=c(2014,1),frequency=12)

plot(sales_ts, main="Monthy Sales Data(Jan 2014 - Dec 2016)",xlab="Time", ylab="Sales", col="blue", lwd=2)
```



Python Programs

1. WAP to swap two numbers

```
num1 = int(input("Enter first number: "))

num2 = int(input("Enter second number: "))

print()

print("Before swap")

print("Number1 = " , num1, " & Number2 = " , num2)


temp = num1

num1 = num2

num2 = temp


print()

print("After swap")

print("Number1 = " , num1, " & Number2 = " , num2)
```

```
● python-programs-sem3 › python -u "c:\Code\Python\python-programs-sem3\one.py"

Enter first number: 7

Enter second number: 0

Before swap
Number1 =  7  & Number2 =  0

After swap
Number1 =  0  & Number2 =  7
```

2. WAP to check whether the given number is even or odd using conditional statement.

```
num = int(input("Enter a number: "))
if num % 2 == 0 :
    print(num, "is even")
else:
    print(num, "is odd")
```

```
python-programs-sem3 python -u "c:\Code\Python\python-programs-sem3\two.py"
Enter a number: 777777
777777 is odd

parih  python-programs-sem3 ✓                                    in pwsh
```

3. WAP to check whether the given year is leap year or not using conditional statement.

```
year = int(input("Enter the year to check : "))
if (year % 400 == 0) and (year % 4 == 0) or (year % 100 != 0):
    print(year, "is a leap year")
else:
    print(year, "is not a leap year")
```

```
python-programs-sem3 python -u "c:\Code\Python\python-programs-sem3\three.py"
Enter the year to check : 2024
2024 is a leap year

parih  python-programs-sem3 ✓                                    in pwsh
```

4. WAP to find max. number from list of no's using loops

```
numbers_list = [33,22,45,67,89,9,1,23,100]
maxNum = numbers_list[0]
print("The list is", numbers_list,)
for num in numbers_list:
  if num > maxNum:
    maxNum = num
print("The maximum number in the list is :", maxNum)
```

```
● python-programs-sem3 › python -u "c:\Code\Python\python-programs-sem3\four.py"
 The list is [33, 22, 45, 67, 89, 9, 1, 23, 100]

 The maximum number in the list is : 100
○ parih › python-programs-sem3 › ✓
```

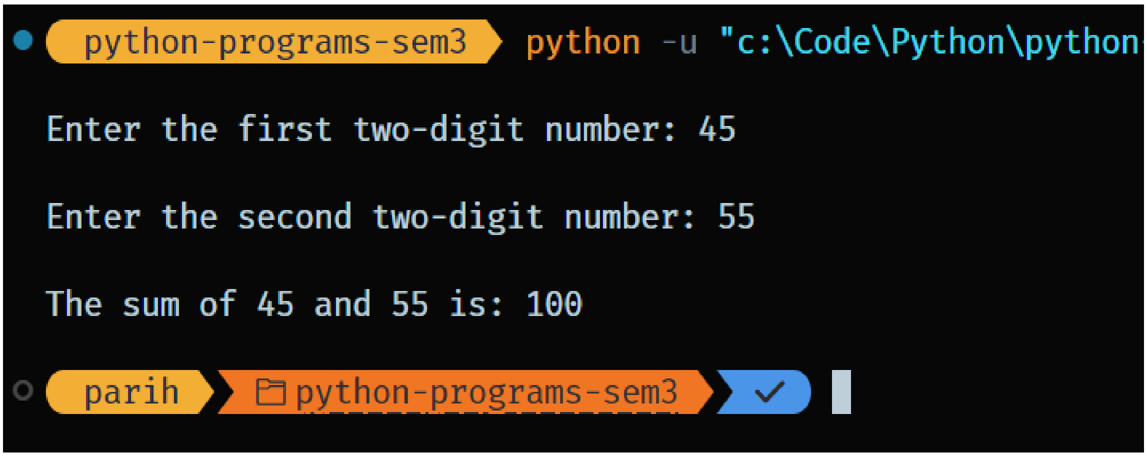
5. WAP to calculate sum of even no's from 1 to 100 including 1 & 100 using range function.

```
start = 1
end = 100
sum = 0
for num in range(start, end + 1, 2):
  sum += num
print("The sum of even numbers from 1 to 100 is:", sum)
```

```
● python-programs-sem3 › python -u "c:\Code\Python\python-programs-s
 The sum of even numbers from 1 to 100 is: 2550
○ parih › python-programs-sem3 › ✓
```

6. WAP to add sum of two-digit numbers.

```
def checkTwoDigit(num):
  if num < 10 or num > 99:
    print("Please enter a valid two-digit number.")
    return False
  return True
first_number = int(input("Enter the first two-digit number: "))
second_number = int(input("Enter the second two-digit number: "))

if (checkTwoDigit(first_number) and checkTwoDigit(second_number)):
  print(f"The sum of {first_number} and {second_number} is: {first_number + second_number}")
else: print("Both numbers should be two-digit numbers.")
```



7. WAP to convert variables from one datatype to another.

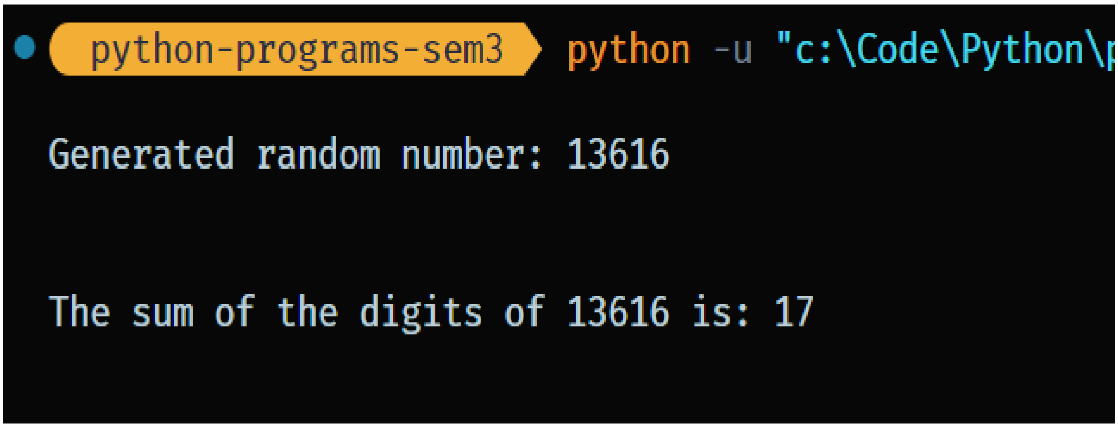
```
def type_conversion_demo():
  int_var = 10
  float_var = 20.5
  str_var = "30"
  converted_float = float(int_var)
```

```python
    print(f"Converted int to float: {converted_float} (Type:    {type(converted_float)})")
    converted_int = int(float_var)
    print(f"Converted float to int: {converted_int} (Type: {type(converted_int)})")
    converted_str_to_int = int(str_var)
    print(f"Converted string to int: {converted_str_to_int} (Type:{type(converted_str_to_int)})")
    converted_int_to_str = str(int_var)
    print(f"Converted int to string: '{converted_int_to_str}' (Type: {type(converted_int_to_str)})")
    str_float = "45.67"
    converted_str_to_float = float(str_float)
    print(f"Converted string to float: {converted_str_to_float} (Type: {type(converted_str_to_float)})")
    list_var = [1, 2, 3]
    converted_list_to_str = str(list_var)
    print(f"Converted list to string: '{converted_list_to_str}' (Type: {type(converted_list_to_str)})")
    converted_str_to_list = eval(converted_list_to_str)
    print(f"Converted string back to list: {converted_str_to_list} (Type: {type(converted_str_to_list)})")
type_conversion_demo()
```

```
python-programs-sem3 > python -u "c:\Code\Python\python-progran

Converted int to float: 10.0 (Type: <class 'float'>)
Converted float to int: 20 (Type: <class 'int'>)
Converted string to int: 30 (Type: <class 'int'>)
Converted int to string: '10' (Type: <class 'str'>)
Converted string to float: 45.67 (Type: <class 'float'>)
Converted list to string: '[1, 2, 3]' (Type: <class 'str'>)
Converted string back to list: [1, 2, 3] (Type: <class 'list'>)
```

8. WAP to generate random numbers and then calculate its sum of digits.

```
import random
def sum_of_digits(number):
  total = 0
  while number > 0:
    digit = number % 10
    total += digit
    number //= 10
  return total
random_number = random.randint(1, 99999)
print(f"Generated random number: {random_number}")
digit_sum = sum_of_digits(random_number)
print(f"The sum of the digits of {random_number} is: {digit_sum}")
```

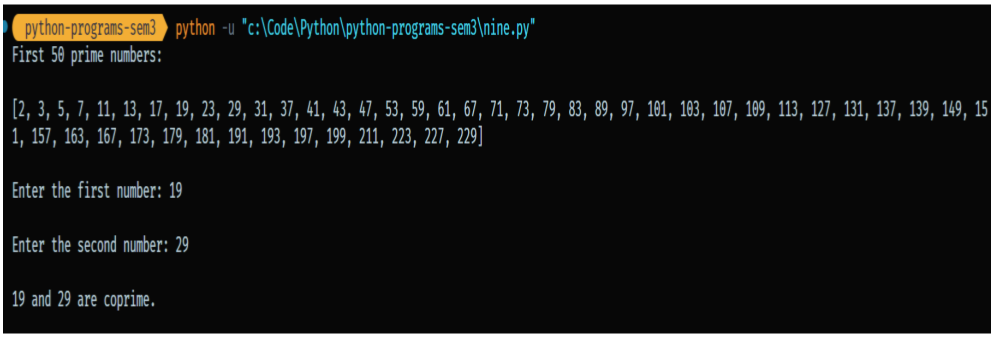


9. WAP to generate first 50 prime number series. Also check whether given any two no's are coprime or not.

```
import math
def is_prime(n):
  if n <= 1:
    return False
  for i in range(2, int(math.sqrt(n)) + 1):
```

```
        if n % i == 0:
            return False
    return True

def generate_primes(count):
    primes = []
    num = 2
    while len(primes) < count:
        if is_prime(num):
            primes.append(num)
        num += 1
    return primes

def gcd(a, b):
    while b:
        a, b = b, a % b
    return a

def are_coprime(a, b):
    return gcd(a, b) == 1

first_50_primes = generate_primes(50)
print("First 50 prime numbers:")
print(first_50_primes)
num1 = int(input("Enter the first number: "))
num2 = int(input("Enter the second number: "))

if are_coprime(num1, num2):
```

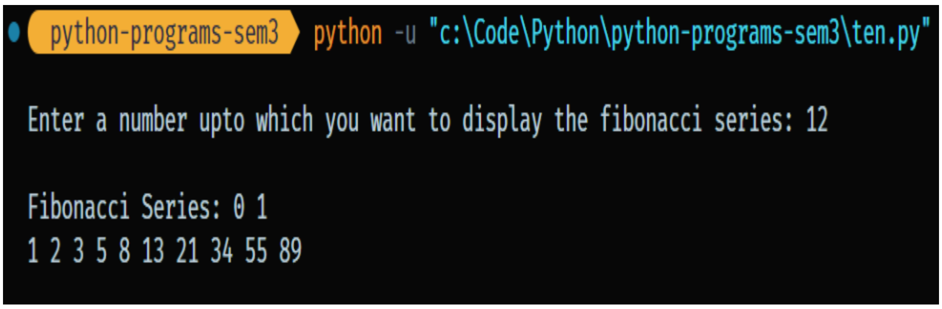
```
    print(f"{num1} and {num2} are coprime.")
else:
    print(f"{num1} and {num2} are not coprime.")
```



```
python-programs-sem3 python -u "c:\Code\Python\python-programs-sem3\nine.py"
First 50 prime numbers:

[2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, 101, 103, 107, 109, 113, 127, 131, 137, 139, 149, 15
1, 157, 163, 167, 173, 179, 181, 191, 193, 197, 199, 211, 223, 227, 229]

Enter the first number: 19

Enter the second number: 29

19 and 29 are coprime.
```

10.WAP to display Fibonacci series.

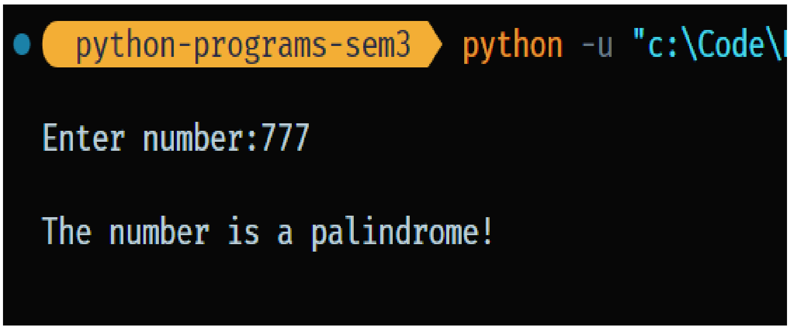
```
num =  int(input("Enter a number upto which you want to display the fibonacci series: "))
n1, n2 = 0, 1
print("Fibonacci Series:", n1, n2, end=" ")
for i in range(2, num):
    n3 = n1 + n2
    n1 = n2
    n2 = n3
    print(n3, end=" ")
print()
```



```
python-programs-sem3 python -u "c:\Code\Python\python-programs-sem3\ten.py"

Enter a number upto which you want to display the fibonacci series: 12

Fibonacci Series: 0 1
1 2 3 5 8 13 21 34 55 89
```

11.WAP to check whether a number is palindrome or not.

```
n=int(input("Enter number:"))
temp=n
rev=0
while(n>0):
   dig=n%10
   rev=rev*10+dig
   n=n//10
if(temp==rev):
   print("The number is a palindrome!")
else:
   print("The number isn't a palindrome!")
```



12. WAP to perform the following operations on string:

a. find index of particular character.

```
text = input("Enter the string: ")
char_to_find = input("Enter the character to find: ")

if char_to_find in text:
   index = text.index(char_to_find)
   print(f"The index of '{char_to_find}' is: {index}")
else:
print(f"'{char_to_find}' not found in the string.")
```

```
python-programs-sem3 python -u "c:\Code\Python\
Enter the string: Machine Learning using Python
Enter the character to find: P
The index of 'P' is: 23
```

b. perform string uppercase and lowercase.

```
print("String in lowercase:", text.lower())
print("String in uppercase:", text.upper())
```

```
Enter the string: Python
String in lowercase: python
String in uppercase: PYTHON
```

c. count the occurrence of particular character in string.

```
text = input("Enter the string: ")
char = input("Enter the character to find and count: ")
if char in text:
   index = text.index(char)
   print(f"The index of '{char}' is: {index}")
else:
   print(f"'{char}' not found in the string.")
count = text.count(char)
print(f"Count of '{char}' in '{text}': {count}")
```

```
 parih  python-programs-sem3  ✓  python -u "c:\(
on\python-programs-sem3\twelve.py"

Enter the string: Machine Learning using Python

Enter the character to find and count: n

The index of 'n' is: 5

Count of 'n' in 'Machine Learning using Python': 5
```

13. WAP to perform operations on sets.

```
s1 = {1, 2, 3, 4, 5}

s2 = {4, 5, 6, 7, 8}

print("Union of sets: ", s1.union(s2))

print("Intersection of sets: ", s1.intersection(s2))

print("Symmetric Difference: ", s1.symmetric_difference(s2))

print("Are sets disjoint: ", s1.isdisjoint(s2))
```

```
ms-sem3\thirteen.py"

Union of sets:  {1, 2, 3, 4, 5, 6, 7, 8}

Intersection of sets:  {4, 5}

Symmetric Difference:  {1, 2, 3, 6, 7, 8}

Are sets disjoint:  False
```

14. WAP to perform Operations on Tuples.

```
x = ('a', 'b', 'c', 'b')

y = ('d', 'e', 'f')

print("Count of b: ", x.count('b'))

print("Position of c: ", x.index('c'))
```

```
z = x+y
print("Joining two tuples:", z)
l = list(x)
print(type(l))
```

```
Count of b:  2

Position of c:  2


Joining two tuples: ('a', 'b', 'c', 'b', 'd', 'e', 'f')

<class 'list'>
```

15. WAP to perform operations on List.

```
list1 = [1, 2, 3, 4, 5, 6]
list2 = [7, 8, 9, 10, 11]
print("Joining two lists: ", list1+list2)
list2.append(12)
print("After appending: ", list2)
list1.reverse()
print("After reversing: ", list1)
x = [3, 15, 4, 1, 2]
x.sort()
print("After Sorting: ", x)
print("Maximum element of list: ", max(list1))
```

```
Joining two lists:  [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]

After appending:  [7, 8, 9, 10, 11, 12]

After reversing:  [6, 5, 4, 3, 2, 1]

After Sorting:  [1, 2, 3, 4, 15]
Maximum element of list:  6
```

16. WAP for Exception Handling.

```
try:
  num1 = int(input("Enter the numerator: "))
  num2 = int(input("Enter the denominator: "))
  result = num1 / num2
  print(f"The result is: {result}")
except ZeroDivisionError:
  print("Error: Division by zero is not allowed.")
except ValueError:
  print("Error: Invalid input. Please enter numeric values.")
except Exception as e:
  print(f"An unexpected error occurred: {e}")
finally:
  print("Program execution completed.")
```

```
Enter the numerator: 7
Enter the denominator: 0
Error: Division by zero is not allowed.
Program execution completed.
```

17. WAP to demonstrate Naïve Bayes.

```
!pip install pandas
!pip install numpy
!pip install scikit-learn
import pandas as pd
import numpy as np
import sklearn
import os

cwd = os.getcwd()
print(cwd)
dataset=pd.read_csv('heart.csv' , engine='python')
with open('heart.csv') as fp:
        data = fp.read().split('\n')

print(dataset)
```

```
      age  sex  cp  trestbps  chol  fbs  restecg  thalach  exang  oldpeak  \
0      52    1   0       125   212    0        1      168      0      1.0
1      53    1   0       140   203    1        0      155      1      3.1
2      70    1   0       145   174    0        1      125      1      2.6
3      61    1   0       148   203    0        1      161      0      0.0
4      62    0   0       138   294    1        1      106      0      1.9
...   ...  ...  ..       ...   ...  ...      ...      ...    ...      ...
1020   59    1   1       140   221    0        1      164      1      0.0
1021   60    1   0       125   258    0        0      141      1      2.8
1022   47    1   0       110   275    0        0      118      1      1.0
1023   50    0   0       110   254    0        0      159      0      0.0
1024   54    1   0       120   188    0        1      113      0      1.4

      slope  ca  thal  target
0         2   2     3       0
1         0   0     3       0
2         0   0     3       0
3         2   1     3       0
4         1   3     2       0
...     ...  ..   ...     ...
1020      2   0     2       1
1021      1   1     3       0
1022      1   1     2       0
1023      2   0     2       1
1024      1   1     3       0

[1025 rows x 14 columns]
```

X = dataset.drop('target',axis=1)

Y = dataset['target']

from sklearn.model_selection import train_test_split

X_train, X_test,Y_train, Y_test = train_test_split(X,Y, test_size = 0.30)

from sklearn.naive_bayes import MultinomialNB

classifier=MultinomialNB()

classifer.fit(X_train,Y_train)

```
▾ MultinomialNB
MultinomialNB()
```

print(classifier.predict(X[:]))

```
[1 0 0 ... 0 1 0]
```

```
Y_pred=classifier.predict(X_test)
from sklearn.metrics import accuracy_score

print (accuracy_score(Y_test,Y_pred))
```

```
0.7435064935064936
```

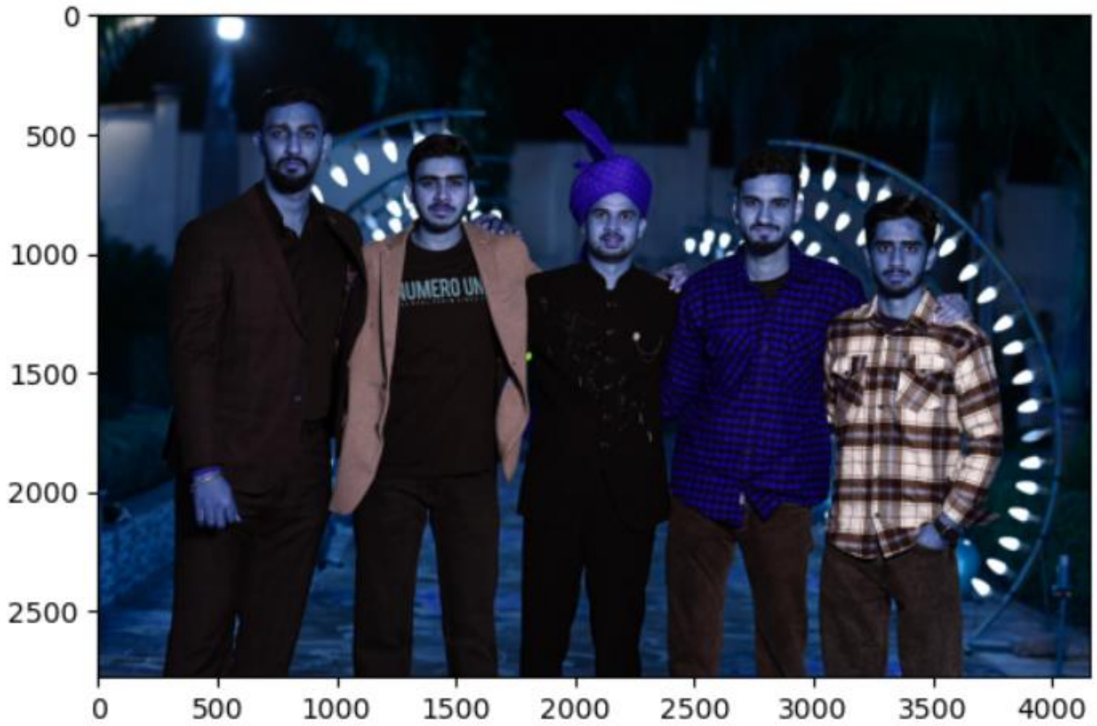
```
from sklearn.metrics import confusion_matrix

cm=confusion_matrix(Y_test,Y_pred)
print(cm)
```

```
[[105  43]
 [ 36 124]]
```

18. WAP to demonstrate CNN for edge detection in Conv2D.

```
!pip install tensorflow

import tensorflow as tf

!pip install keras

from tensorflow import keras

!pip install matplotlib

import matplotlib.pyplot as plt

import numpy as np

!pip install opencv-python

import cv2
img = cv2.imread("mlx.jpg")
print(img)

plt.imshow(img)
```

<matplotlib.image.AxesImage at 0x1ea008f4560>



```
from tensorflow.keras import layers
from tensorflow.keras.layers import Conv2D
newImg=Conv2D(filters=3,strides=(1,1),kernel_size=(3,3),padding="same"
,use_bias=False)
print(img.dtype)
print(img.shape)
img=tf.cast(img,tf.float32)/255
kernel=np.array([[0,-1,0],[-1,4,-1],[0,-1,0]],dtype=np.float32)
print(kernel.dtype)
img=np.expand_dims(I,axis=0)
print(img.shape)
print(kernel.shape)
kernel = kernel[:,:,np.newaxis,np.newaxis]
print(kernel.shape)
kernel = np.repeat(kernel,repeats=3,axis=2)
kernel=np.repeat(kernel,repeats=3,axis=3)
print(kernel.shape)
newImg.build(input_shape=img.shape)
newImg.set_weights([kernel])
```

```
output=newImg(img)[0]
plt.imshow(output)
```

<matplotlib.image.AxesImage at 0x1ea009b6cc0>